



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

B.TECH- V SEM

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

W.E.F JUNE 2020

S.No.	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hour s per week			Total Credits
			Theory			Practical			L	T	P	
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem	Term work					
							Lab Work & Sessional					
1.	EC 51	MICROPROCESSOR & ITS APPLICATION	70	20	10	30	20	150	3	1	2	5
2.	EC 52	DIGITAL COMMUNICATION	70	20	10	30	20	150	3	1	2	5
3.	EC 53	COMMUNICATION NETWORK AND TRANSMISSION LINES (CNTL)	70	20	10	30	20	150	3	1	2	5
4.	EC 54	<u>ELECTROMAGNETIC THEORY</u>	70	20	10	-	-	100	3	1	0	4
6.	EC 55	MATLAB PROGRAMMING	-	-	-	30	20	50	-	-	2	2
7.	EC-56	MINOR PROJECT 1	-	-	-	-	100	100	-	-	2	2
9.	EC 57	SEMINAR	-	-	-	-	50	50	-	-	1	1
		Total	280	80	40	120	230	750	12	4	8	24



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SYLLABUS

EC-51 MICROPROCESSOR AND ITS APPLICATIONS

UNIT I

Salient features of advanced microprocessors. RISC & CISC processors. Review and evolution of advanced microprocessors: 8086, 8088, 80186/286/386/486/Pentium, introduction to 8086 processor: Register organization of 8086, Architecture, signal description of 8086, minimum mode 8086 systems and timings and maximum mode 8086 systems and timings

UNIT II

Intel 8086 microprocessor programming: 8086 Instruction Set, Addressing modes, Assembly Language Programming with Intel 8086 microprocessor

UNIT III

Introduction to the various interfacing chips like 8155, 8255, Interfacing key boards, LEDs, ADC, DAC and memory Interfacing.

UNIT IV

General purposes programmable peripheral devices (8253), 8254 programmable interval timer, 8259A programmable interrupt controller & 8257 DMA controller, USART, serial I/O & data Communication.

UNIT V

Introduction to microcontrollers (8051) and embedded systems: 8051 architecture, pin description, I/O configuration, interrupts, addressing modes, an overview of 8051 instruction set, embedded system, use of microcontrollers in embedded systems

Reference Books:

1. Advance microprocessor and peripheral – A.K. Ray and K. M. Bhurchandi, Tata Mcgraw Hill
2. Microprocessor and Interfacing – D.V. Hall, McGraw Hill.
3. The Intel microprocessor - Barry B. Brey, Pearson
4. The 8086 & 8088 Microprocessor- LIU and Gibson, Tata McGraw Hill
5. The 8051 microcontroller and embedded systems-M.A. Mazidi, Janice Gillispie Mazidi, Pearson Prentice Hall

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EC-52 DIGITAL COMMUNICATION

Unit I

Sampling theorem for low pass and band pass signals, Ideal sampling, Natural sampling, Flat top sampling, crosstalk, aliasing, time division multiplexing, PAM, PWM and PPM their generation and detection.

Unit II

Pulse code modulation, Quantization, quantization noise, companding, Inter symbol interference, Eye pattern, Delta and adaptive modulation, Encoding techniques: On-Off signaling, Polar signaling, RZ signaling, Bipolar signaling, AMI, Manchester code, Differential encoding their advantage and disadvantages.

Unit III

Band pass data transmission: ASK, Binary phase shift keying (BPSK), QPSK, DPSK, coherent and non coherent BFSK, minimum shift keying, QAM, Concept of M-ary PSK and M-ary FSK. Spectral properties of QPSK and MSK.

UNIT IV

Matched filter and correlator detector. Gram Schmidt orthogonalization procedure and concept of signal space for the computation of probability of error, calculation of error probability for BPSK, QPSK, QAM and coherent BFSK, comparison of different modulation techniques.

Unit V

Concept of information theory, entropy, information rate, channel capacity, Shannon's theorem, Shannon Hartley theorem, BW and signal to noise ratio trade off, sources encoding, extension of zero memory source, Error correcting codes: linear block codes and cyclic codes: encoder and decoder circuits, burst error correcting codes, concept of convolution codes.

Reference Books:

1. Communication Systems –Simon Haykins, Wiley
2. Principle of Communication Systems-Taub and Schilling, Tata McGraw-Hill
3. Communication Systems-Singh and Sapre, Tata McGraw-Hill

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EC- 53 Communication Network and Transmission Lines (CNTL)

Unit I

Characteristic Parameters of symmetrical and asymmetrical two port networks and their design Image impedance, iterative impedance, characteristic impedance, propagation coefficient, image transfer coefficient, iterative transfer coefficient, Lattice and Bridged T networks, reactive matching networks, matching techniques, insertion loss, symmetrical and asymmetrical attenuators and their design.

Unit II

Passive LC Filters Analysis and design of Low pass, high pass, band pass and band elimination filters, m-derived filters, composite filters, Filter specifications, Butterworth approximation, Chebyshev approximation, elliptic function approximation, frequency transformation.

Unit III

Positive real function LC, RL, RC, and RLC network synthesis, Foster and Cauer network, minimum positive real function, Brune's method, Bott-Duffin method, Synthesis-Coefficient.

Unit IV

Transmission line fundamentals Lumped parameter equivalent, voltage and current on a transmission line, infinite line, characteristic impedance and propagation constant, waveform distortion, attenuation and phase equalizers, distortion-less line, loading, liner reflection on a line, reflection coefficient, input and transfer impedances, open circuit and short circuit line, reflection factors, reflection loss, insertion loss, T and π equivalents of a line, location of line fault, construction and design of two wire line and coaxial cable.
Academic Session 2017-18

Unit V

Line at radio frequencies Parameters of line and coaxial cable at radio frequencies, dissipation-less line, voltage and current on a dissipation-less line, standing waves, standing wave ratio, input impedance of open circuit and short circuit, power and impedance measurement on lines, eighth-wave, quarter-wave and half wave line, circle diagram, Smith chart, solution of problems using Smith chart, single and double stub matching .introduction to micro-strip lines and its analysis.

References:

1. Ryder: Networks and Transmission Lines, PHI Learning.
2. Valkenberg: Introduction to Modern Network synthesis, Wiley India.
3. Suresh: Electric Circuits and Networks, Pearson Education.
4. Raju: Electromagnetic field theory and Transmission Lines, Pearson Education.
5. Ganesan: Transmission Lines and Waveguides, TMH.
6. Rao: Electromagnetic Waves and Transmission Lines, PHI learning

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EC- 54 ELECTROMAGNETIC THEORY

Unit I

Steady Electric Field: Coulomb's Law, units, Electric field intensity, Electric flux and flux density, Gauss law, Boundary relations, concept of divergence, Curl, scalar and vector potential. electric field in dielectric and conductor, continuity equation, methods of images.

Unit II

Magnetic field due to steady currents, force between current carrying wires, Stokes theorem, vector magnetic potential, magnetization vector and its relation to magnetic field.

Unit III

Maxwell's Equation: Time varying field and displacement current, faraday's law.

Unit IV

Wave Equation: Pointing vector, Plane electromagnetic waves in free space, dielectric medium and conducting medium, Skin depth, slepian vector.

Unit V

Waves propagation in lossy dielectrics, plane waves in lossless dielectrics, reflection of a plane wave at normal incidence , reflection of a plane wave at oblique incidence .

Reference Books:

1. Elements of Engineering Electromagnetic Third Edition- N.N. Rao- Prentice Hall, India.
2. Elements of Electromagnetic, Second Edition- Matthew N.O. Sadiku- Saunders coll Publishing.
3. Fields & Waves in Communication Electronics- S.Ramo, J.R. Whinnery & T. Van Duzer- John Wiley & Sons.
4. Electromagnetic- J.D. Kraus-McGraw Hill
5. Electromagnetic Waves & Radiating Systems- E.C. Jordan & K.G. Balmain- Prentice Hall.

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EC-55 MATLAB PROGRAMMING

1. Study of Introduction to MATLAB

2. Study of basic matrix operations

3. To solve linear equation

4. Solution of Linear equations for Underdetermined and Over determined cases

5. Determination of Eigen values and Eigen vectors of a Square matrix.

6. Solution of Difference Equations.

7. Solution of Difference Equations using Euler Method.

8. Solution of differential equation using 4th order Runge- Kutta method.

9. Determination of roots of a polynomial.

10. Determination of polynomial using method of Least Square Curve Fitting

11. Determination of polynomial fit, analyzing residuals, exponential fit and error bounds from the given data.

12. Determination of time response of an R-L-C circuit.

EC-56 MINOR PROJECT

EC-57 SEMINAR



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